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Research article

Sustainability with environmental policy stringency and financial development for green technological innovations: Evidence from Sub-Saharan Africa

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A R T I C L E I N F O

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A B S T R A C T

This study is prompted by the United Nations Sustainable Development Goals (UN-SDGs) and their anticipated effect by 2030. Financial development and environmental policy stringency are becoming crucial tools for tackling environmental degradation. However, there is a limited body of research that explores their interactive roles for green technological innovation and contributes to environmental sustainability, particularly in Sub-Saharan African economies. In this study, we investigated how financial development and environmental policy stringency moderate the relationship between technical innovation and load capacity factor in the panel of 29 Sub-Saharan African (SSA) countries from 1990 to 2020. The study employed multifaceted empirical techniques, including Dynamic Ordinary Least Square (DOLS), Fully Modified Ordinary Least Square (FMOLS), Canonical Cointegration Regression (CCR), and method of Moment Quantile Regression (MMQR) estimators. The findings show that coefficients of the interaction of financial development and green innovation (FD*INV), environmental policy stringency and green innovation (EPS*INV), and financial development and energy transition (FD*ET) indicate a positive contribution to environmental sustainability. Further, feasible generalized least squares (FGLS) and bootstrap quantile regression (BSQR) approaches are used to check the robustness of the results. This study offers a policy framework that focuses on developing a sustainable economy to achieve SDGs 7 & 13. The primary objective is to build a green economy and ensure long-term environmental sustainability in SSA countries.

1. Introduction

The environmental catastrophe surpasses national boundaries. Currently, unsustainable manufacturing and consumption patterns are the underlying causes of three critical issues; loss of biodiversity, global warming, and environmental degradation. Rising global warming is a global challenge, becoming a matter of great concern for all economies and policymakers around the globe. All countries are apprehensive for sustainable development owing to the continuous environmental degradation (Menne et al., 2020; van Asselt and Green, 2023). Due to detrimental effects of environmental contamination on economy and society, the United Nations has established the Sustainable development goals (SDGs), aims to alleviate the devastating effects of fossil fuels extractions, focusing on sustainable development through threefold bottom-line framework; economic, social, and environmental dimensions. SDG 13 intends to implement appropriate and effective strategies to address climate disruptions and its consequences. It is significantly correlated with remaining 16 SDGs of 2030 agenda for sustainable development. In response to rising global warming, countries globally endorsed the Paris Agreement at COP21 and SDGs 2030 agenda, attempting to restrict increase in global temperature to a level below 2^{o} C, considering significant challenges, to strive for an aggregate rise of 1.5^{o} C (Kwakwa et al., 2021; Lu and Wang, 2024). These circumstances portray a bleak illustration of hazardous consequences of

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contaminants on environment, necessitating an investigation of primary causes contributing to rising emissions for determining and redesigning environmental policies. Further, policymakers and academicians need to develop an understanding to effectively address environmental challenges through effective policy responses.

Africa is the 2nd most susceptible region to global warming and has the least level of global climate readiness (Shulla et al., 2021; Martín--Blanco et al., 2022). The indicators of environmental fragility and global warming readiness reveal significant disparities among various economies. Sub-Saharan Africa (SSA) has restricted economic growth trends, generally characterized with insufficient production and technologically driven businesses. Furthermore, many studies (Razzaq et al., 2021; Wang et al., 2023a, 2023b; Bradu et al., 2023) highlighted that the economic system of SSA have less diversification than other developed nations. This lack of diversity may hinder the adoption of green technologies and sustainable conduct. The lack in economic diversity results in persistent dependency on conventional and resource-intensive industries, consequently increasing environmental degradation. The adverse effects of inadequate economic diversity transcend the economic domain, impacting SSA potential to shift towards sustainable industrial practices. Furthermore, SSA experiences a technological disparity and an impairment in innovation. Advancements in technologies, crucial for sustainable growth, progress at a slower pace in SSA region. This technological deficiency hinders the deployment of green innovative technologies, putting the region vulnerable to obsolete and environmentally destructive methods. Limited funding in R&D, inconsistent adoption of innovative technologies, slower energy transition possesses a challenge for SSAregion to synchronize its economic activities with sustainable practices. Consequently, SSA region encounters challenges in mitigating adverse environmental effects of economic activities and shift towards environmentally sustainable growth trajectories. Therefore, the current study seeks to expand on the rationale of transition to renewable energy sources and technological innovation and financial development as a potential strategy to accomplish environmental sustainability and SDG 13 in SSA region.

Moreover, the premise of financial development and economic development constitutes the basis for research on the association of technical innovation and financial development (Ramzan et al., [illegible]). Through the processes of collecting data and evaluating risk variability, and lending restriction, the financial industry contributes to the persistence of technical innovation initiatives and the enhancement of technological innovation efficacy (Wen et al., [illegible]; Fatima et al., 2023a). However, the financial system in SSA economies is the least developed globally. Allen et al. (2011) contends that before the global financial crisis, the financial intermediaries and credit availability in SSA remained significantly inadequate compared to those in other developing regions. For instance, in SSA, the average proportion of credit given to the private sector as a percentage of GDP was 17%, whereas in other emerging regions, this ratio ranged from 32% to 43%. SSA's financial sector has made some recent advancements, although it remains the least developed compared to other developing regions (Taddese Bekele and Abebaw Degu, 2023). Hence, comprehending the influence of financial development on technological innovation in SSA region has significance for sustainable development and climate change strategies (Liu et al., 2022). Therefore, if it is determined that there is a significant association between financial development and an increase in environmental sustainability, it will be more feasible to accomplish SDGs in SSA region. Conversely, if it is observed that financial development leads to degrade the environment, this might have an impact on environmental sustainability forecasting models and climate change policy in SSA region. The financial policymakers of the UN Environment Program Finance Initiative (2022), emphasized the significance of green investment in green technologies and energy transition for sustainable development in SSA region. The present study seeks to address this policy void.

Environmental degradation generally arises from distinctive conduct of human activities which is primarily destructive (Hou et al., 2023; Khan et al., 2024). It is crucial to comprehend drivers of environmental degradation to formulate effective environmental policies. SSA region is not only abundant in resources yet heavily reliant on them, increasing the extraction and combustion activities which upsurge environmental degradation. However, implementation of strict environmental regulations, the preservation of natural resources and tackling pollution could significantly affect sustainability in the environment. Implementation of strict environmental policies ensures adequate and sustainable consumption of natural resources, hence eliminating negative environmental consequences. Therefore, considering policy context, SSA countries experienced challenges in maintaining environmental quality through emission mitigation and abatement strategies. Environmental Policy Stringency (EPS) has always been a significant aspect of encouraging green innovative technologies but their efficacy is still uncertain without encouragement in SSA region. EPS has the potential to lower the adverse consequences of environmental deterioration promoting green innovative technologies and discouraging environmentally inadequate and emission-intensive practices. Consequently, EPS and environmental taxes can influence manufacturing and consumption patterns towards environmentally sustainable practices and the manufacturing of energy-intensive commodities. EPS is rapidly becoming a fundamental driver of sustainable ecosystem and an effective approach for cutting emissions. However, the current study emphasizes this approach as one of the effective measures available to mitigate the detrimental consequences of increasing global temperature and environmental degradation.

Therefore, these significant discussions emphasize the necessity of investigating the issue in a more comprehensive way within the context of SSA countries. This examination involves the utilization of various socio-economic and environmental factors to assess the United Nations' SDGs in terms of conserving resources and increasing environmental sustainability. For this purpose, the SSAcountries need to realign environmental policies and green funds to encourage green innovative technologies to attain environmental sustainability. However, the redesigning and reformulation of these environmental policy factors must integrate the adverse effects that are triggered by energy use pattern practices while simultaneously improving positive effects of implementing stringent environmental policy and financing renewable energy projects. Therefore, this justifies the formulation of policy structure that reforms environmental policies through implementation of green innovative technologies to achieve environmental sustainability. This argument supports the significance of policy framework in SSAregion to attain SDGs 7 & 13 through redesigning environmental policies. This policy discourse establishes following research questions that also reorient the policy agenda to achieve 2030 goals.

i. To what extent environmental policy stringency, green innovation, energy transition, financial development and economic growth on the load capacity factor in SSA countries?
ii. Does financial development moderate the association of green innovation and load capacity factor in SSA countries?
iii. How does financial development moderate Energy Transition for environmental sustainability in the SSA region?
iv. How does environmental policy stringency moderate the association of green innovation and load capacity factor in SSA countries?
v. Does EPS and INV statistics confirm Porter Hypothesis in SSA region?

The significance of policy level optimization of green innovative technologies is apparent in the discussion of environmental sustainability and its drivers in SSA region. This policy framework suggested in the current study is intended to facilitate and promote the achievement of SDGs 7 & 13 by SSA countries. In pursuing this objective, the current study analyzes the determinants of sustainability of environment and

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moderating role of EPS and FD for 29 SSA countries for the period of 1990–2020. This current study offers a fresh SDGs pronged policy structure for promoting green innovative technologies and energy transition by implementing stringent environmental policies and green financing for environmental sustainability. Fig. 1 demonstrates that the Sustainable Africa Scenario (SAS) highlighted investment solutions in all over the SSA region that are crucial for encouraging energy investment. The cost of financing for renewable energy initiatives in SSA region is 2–3 times more than other developed countries, hindering investment by raising costs associated with renewable energy projects.

According to World Energy Outlook Report (2023), the significance of growing energy investments is to surpass 200 billion USD by 2030. These goals are crucial for SSA countries to fulfill their renewable energy development goals. To accomplish SDG 7 SSA economies must shift 130 million people from consumption of fossil fuels. To achieve SDGs and ensuring easy access to renewable energy sources, it is crucial to tackle energy investment void in SSA region. Realizing renewable energy promise and promoting green energy future for SSA region necessitates collaborations and redesigning policies. The current study fills this policy void.

These above objectives required an adequate methodology and analysis of this construct. These aspects will strengthen proposed policy structure. In this regard, governments in SSA nations are implementing policies and programs to promote energy transition and technological innovation to enhance environmental sustainability. This supports the validity of the Porter Hypothesis. This research makes significant contributions to the existing literature, consistent with the rationale for this research. First, the current study provides insights into how FD moderates the effectiveness of energy transition (ET*FD), and green technological innovation (INV*FD) in improving environmental sustainability (LCF) in the SSA region. Secondly, this research demonstrates the effect of interaction of EPS and green innovation (INV*EPS) on LCF. Therefore, in contrast to earlier research, this study enhances the existing body of knowledge by investigating the influence of financial development and EPS as moderators on the relationship between energy transition, green innovation, and LCF in SSA economies. By analyzing these associations this study contributes to understanding the synergies between FD, ET, EPS, and INV and offer policies for achieving SDGs (7 & 13). [illegible] study employs extensive econometric techniques [illegible]OLS, FMOLS, CCR and MMQR on panel dataset to analyze the impacts of selected variables on environmental sustainability. Further, this study also employed FGLS and BSQR approaches to check the robustness of the primary models of the study.

The remainder is structured as follows. Section 2 discusses the theoretical basis and development of ideas relating to sustainability. Section 3 discusses the data and approach. Section 4 summarises and examines the empirical results, followed by policy suggestions in Section 5.

2. Literature review

The objective of green technology innovation is to mitigate pollution, enhance the effectiveness of technological advancements, and promote the simultaneous development of environmental preservation and economic development. The implementation of this notion incorporates two crucial aspects: technological innovation and sustainable development. It is imperative to enhance input-output efficacy and economic gains simultaneously executing the fundamentals of green growth. The influence of financial development on technology innovation is significant, however, environmental regulation primarily drives the growth of innovative green technologies. This research investigates the moderating role of the advancement of financial systems and the implementation of strict environmental policies in driving the development of green innovative technologies.

2.1. *Financial development, green technological innovation and environmental sustainability*

Financial development incorporates various dimensions, such as modifications in the structure of the financial system (Erdoğan et al., 2020), the magnitude of financial activities, and the efficacy of financial operations ([illegible] et al., 2021; Hossain et al., 2023). The integration of these three factors can provide a more thorough reflection of the quality of financial development ([illegible] et al., 2023). The financial system facilitates the investment of funds into technological research and development efforts of businesses (Jiakui et al., 2023; Cheratian et al., 2024), consequently promoting and incentivizing their innovative R&D pursuits ([illegible]bdin et al., 2024). Due to the risk-averse nature of the financial industry, the evaluations for technological innovation initiatives are rigorous. However, the financial market has a stronger propensity to [illegible] technology research and development activities that entail substantial risks but offer commensurate rewards (Andrei et al., 2024). Consequently, it is prepared to allocate capital loans to technology R&D activities. The effect of both factors on innovation production varies (Smith et al., 2024). Increasing the financial resources allocated to R&D can enhance the funding available for technological innovation (Zhu et al., 2020). This, consequently, may boost the resilience of R&D efforts and contribute to increased innovation output (Zheng et al., 2023). Enhancing financial effectiveness can expedite the rotation of funds, address the finance limitations experienced by businesses in their innovation activities, and attain innovative and effective solutions (Gomes and Pinho, 2023).

Green technology innovation has the potential to mitigate ecological environmental contamination, minimize the depletion of natural resources (Akram et al., 2023), and enhance energy sources (Ahakwa et al., 2023) approaches, and solutions (Shi et al., 2024). It can assist businesses in improving their primary competitive advantages and also contribute to the advancement of sustainable total factor productivity and the growth of a global sustainable ecosystem (Feng et al., 2024).

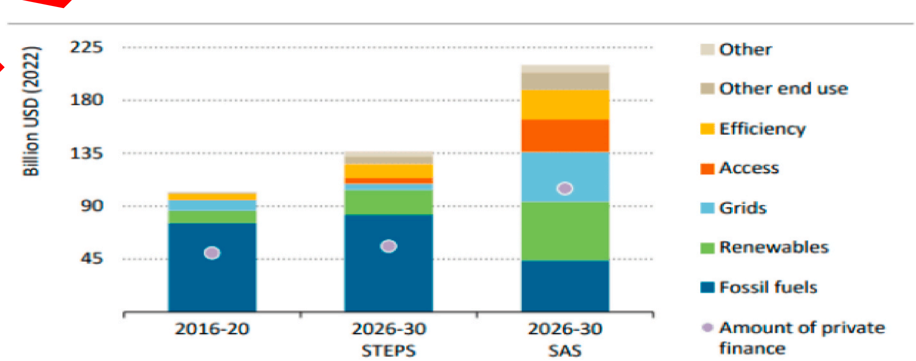

Fig. 1. Annual investment in renewable energy sources in SSA regions, 2016–2030; Source: World Energy Outlook Report (2023).

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Financial development is intricately associated with economic growth and the environment (Saqib et al., 2024), as it stimulates environmental sustainability by exerting influence on technological innovation (Mehmood et al., 2024). Furthermore, Wang et al. (2023a, 2023b) examined the impact of technological advancements, financial development, renewable and non-renewable energy sources, and foreign direct investment (FDI) inflows on the ecological footprint of 14 emerging European Union nations. To accomplish this, the panel data for these countries spanning from 1995 to 2020 is considered by employing second-generation panel econometric tests, specifically the Augmented Mean Group (AMG) and Common Correlated Effects Mean Group (CCEMG) estimators, to investigate the highlighted relationship, taking into consideration cross-sectional dependency and slope heterogeneity. They concluded that renewable energy and technical innovation help reduce environmental degradation, whereas financial development, non-renewable energy use, and foreign direct investment (FDI) contribute to its long-term deterioration. Yang et al. (2021) investigated the influence of remittance inflows, technical advancements, and financial development on the state of the environment in the economies of Brazil, India, China, and South Africa (BICS) from 1990 to 2016. The results of the present study indicate that the influx of remittances and the development of the financial sector have a major negative impact on environmental quality. However, technological improvements play a crucial role in reducing the level of ecological footprint. The study unveiled the inherent process by which financial development, business innovation, and environmental sustainability are interconnected. Furthermore, the advancement of financial systems can mitigate limitations on corporate funding, decrease expenses associated with acquiring information and spreading risks, facilitate the influx of financial resources into technology innovation initiatives, and enhance the efficacy of business technological innovations. FD is crucial [illegible] promoting INV adoption aimed at sustainable development in SS[illegible]gion. Financial institutions provide availability to finances encoura[illegible] investment in INV, energy efficient policies, sustainable practic[illegible] Considering the rising relevance of FD for encouragi[illegible]V for su[illegible]tainable environment in SSA, several research sh[illegible] persist. Inadequate analysis on significance of FD in [illegible]uragin[illegible]NV and limited research on policy and regulatory fra[illegible]rk t[illegible] green financing to achieve environmental [illegible]taina[illegible]n SSA cou[illegible]tries. Addressing these gaps will offer val[illegible] insights [illegible]licymakers, financial institutions and businesses f[illegible] FD as a cat[illegible] for INV, and environmental sustainability w[illegible] ma[illegible] policies and [illegible]vestment decisions in SSA countries.

2.2. Environmental policy [illegible]gency, [illegible]hnological innovation and environmental sustainab[illegible]

Several resea[illegible] have [illegible]erved [illegible] the innovation of green technology m[illegible]ly redu[illegible]egative impact on the natural environme[illegible]rough[illegible] manufac[illegible]ing process and additionally during the co[illegible]ptio[illegible]of a product (Galeotti et al., 2020; Xie et al., 2023)[illegible]otion is [illegible]equently referred to as environmental technology inno[illegible]n. Certain researchers have initiated research on the advancement o[illegible]n technology innovation within the limitations imposed by environmental regulations (Yuan and Zhang, 2020; Shah et al., 2023). Academia holds diverse perspectives regarding the association of environmental regulation and the advancement of green technologies. Several research posit that environmental regulation serves as a motivating aspect for the advancement of green technology innovation (L. J. Liu et al., 2023; Mahajan and Majumdar, 2024). These studies argue that environmental regulation has a positive effect on innovation and can facilitate the adoption of green technological innovation. Lu & Li (2023) analyze the data from 45 publicly traded companies in China's highly polluting sectors, along with panel data from 30 provinces and cities spanning from 2008 to 2017, revealed a significant association between environmental regulations and the development of green technology innovation in various regions of China. This finding supports the validity of the Porter hypothesis. Certain scholars have suggested that environmental regulation hinders the advancement of innovative green technology (Lv et al., 2021; Yu et al., 2023; Farooq et al., 2024). Businesses must allocate resources towards pollution control in order to comply with environmental regulations. However, this form of investment hinders the allocation of resources towards the development of green technologies and amplifies the financial burden on businesses. There is a dearth of consensus among researchers on the impact of environmental regulation and green technology innovation (Bai et al., 2022; Tan et al., 2023). While some companies are considering a restructuring of their internal technology and processes to encourage green economy and promote susta[illegible] (Qiu et al., 2023; Cai, 2023), their primary motivation rem[illegible] pro[illegible]king. However, complying with environmental regula[illegible]s during th[illegible]ansformation may have a negative impact on their fi[illegible]l gains (Wa[illegible] et al., 2023; Naqvi et al., 2023). Consequentl[illegible]e maj[illegible]of busin[illegible]es are unable to actively adhere to the obje[illegible]es of enviro[illegible]ta[illegible]gulations.

Many governments are [illegible]y to en[illegible] pertine[illegible]es and regulations and implement required p[illegible]dur[illegible] regulate the manufacturing and operational activitie[illegible] bus[illegible] (Muir [illegible]., 2023; Krause, 2023; Rauf et al., 2023)[illegible]ironmenta[illegible]ulati[illegible] impose additional costs on small and medi[illegible]d enterprise[illegible]s) for implementing pollution control mea[illegible]s (Jo[illegible] Min, 2023; [illegible]en and Albrecht, 2023; Chithra, 2023). This, in turn, re[illegible]s their ability to invest in competitive activiti[illegible]nnovation. In[illegible]ttempt to mitigate the financial burden (Fe[illegible] al., 2023), businesses will implement strategies to manage the re[illegible]ting ecologic[illegible]degradation (Du et al., 2023), thereby maintaining th[illegible]rmonious f[illegible]tioning of the surrounding ecosystem and sustaining [illegible]sitive pu[illegible] perception (Maspul and Almalki, 2023). Simultaneous[illegible]s frequently encounter challenges when it comes to making substantial investments in the advancement of environmentally [illegible] technologies for processing (Shi et al., 2024). In fact, certain [illegible]es may even shift polluting industries and move away from regions with stringent environmental regulations (Rath et al., 2024). This has a detrimental impact on the vigor of research and development (R&D) and innovation in the area (Lu and Wang, 2024), hindering the advancement of green technologies and impeding the enhancement of innovation effectiveness (Bibri et al., 2024). The Porter hypothesis is an accepted framework for examining the association between environmental regulation and the advancement of green technologies (Ma et al., 2024). Environmental regulations are considered to have an innovation compensation impact, which promotes the advancement of patent technology, the advancement of sustainable technologies, and the adoption of pollution reduction technologies. Several studies demonstrated that EPS encourages INV to achieve environmental sustainability. However, limited research has been conducted on the efficacy of policy instruments in encouraging INV to achieve environmental sustainability specifically in SSA region. By tackling these research limitations policymakers, and businesses might effectively employ EPS to achieve sustainability in SSA, consequently providing access to renewable energy sources and encouraging climate action. Achieving SDGs 7 & 13 requires appropriate policy structures that encourage INV and environmental sustainability. The current study fills this policy void.

2.3. Energy transition and environmental sustainability

The worldwide transition to renewable energy sources is a significant challenge, particularly in addressing environmental issues and achieving long-term sustainability (J. Chen et al., 2023; Zou and Wang, 2024). Currently, there is a growing focus on comprehending the interplay between financial development, advancements in green technology, energy transition and regulations pertaining to the environment (Bogdanov et al., 2021; Guo et al., 2024). This comprehension is crucial for promoting and expediting sustainable development. The contemporary world deteriorates in the absence of a consistent energy source.

The global economy relies on a continuous and abundant supply of energy, while modern society deteriorates (Khan et al., 2022; Balcilar et al., 2023). However, the adoption of non-sustainable techniques to fulfill the energy necessities for the environment has resulted in catastrophic damage. The escalating global temperature resulting from greenhouse gas emissions and the existence of pollutants in the atmosphere and water pose an imminent threat to the well-being of future generations (Hepburn et al., 2021; Zainal et al., 2024). The general recognition of the issue of climate change has significantly elevated since the beginning of the 21st century (Sun et al., 2023; Ofélia de Queiroz et al., 2024). Multiple factions, particularly the general public, advocate for the implementation of risk mitigation strategies pertaining to climate change (Chishti et al., 2023). A significant issue currently on the international political agenda is the transition from traditional to sustainable energy sources (Alola et al., 2023). The energy transition has become a significant effort due to recent worldwide advancements. Internationally, nations are encountering challenges in attaining the objectives of SDGs (Buonomano et al., 2023). The primary obstacle that nations are presently confronting is the transition towards more environmentally friendly energy sources. The necessity for sustainable growth and the advancement of a robust and sound economy relies on both the transition towards more environmentally friendly energy sources and the implementation of sustainable environmental practices. Scant renewable energy targets and subsidies impede transition in the SSA region. Energy demand is further exacerbated by an insufficient renewable energy regulatory framework. Robust policy frameworks that encourage ET, sustainable environment, and climate adaptation are essential to achieve SDGs 7 & 13.

2.4. Literature gap

The above literature serves as a valuable source of inspiration for this research but also possesses certain shortcomings. By far, the literature review reveals the dearth of establishing an extensive policy structure to tackle environmental challenges. Contextualization of policy determinants is crucial for understanding their synergistic effects and there is a void in the literature. EPS and FD could promote the im[illegible] of this policy paradigm. Considering the extent of environmental sustainability in SSA countries, this policy structure could serve as a potential alternative. Although the chall[illegible]gers on a global scale, the dearth of benchmark policy frameworks could impede the achievement of these policy objectives. The policy void illustrates the existing research lacuna in the literature, the current study seeks to address this gap. Further, certain studies employed diverse methodologies to evaluate the level of innovation in green technology, perhaps leading to discrepancies in their measurements. Moreover, while numerous studies have examined the direct association between financial development and technological innovation concerning environmental sustainability, only a few have considered the moderating influence of financial development and the environmental policies on the relationship between green technological innovation and environmental sustainability as well as the moderating role of financial development between energy transition and LCF. Lastly, the literature on environmental regulation lacks consistency in its tool selection, leading to variations in the level of environmental regulation. Finally, previous research relied on CO_2 emissions or ecological footprints as indicators for assessing environmental sustainability. However, the present study employed the Load capacity factor (LCF) to measure environmental sustainability in SSA nations. LCF serves as more integrated indicator of environmental sustainability, because it evaluates the effectiveness of whole energy system. Utilizing LCF enables policymakers to identify potential for promoting energy system effectiveness, eliminating waste and encouraging environmental sustainability in SSA region. Therefore, the present study contributes to the field of environmental studies by expanding upon the literature concerns that have been previously discussed.

3. Methodology

We employ balanced annual panel data of 29 nations in the Sub-Saharan Africa region from 1990 to 2020. This dataset has been diligently collected from multiple sources. The main reason for selecting these SSA nations is that their growth over the SDG timeframe has significantly outperformed both the historical average and the SDG target of 7% per year. Due to the rising population and the substantial influence of COVID-19, which has greatly impeded public discussions on climate change, it is anticipated that the index will continue to decline. Hence the study uses load capacity factor as an explained variable.

3.1. Variables and data sources

Load capacity factor (LCF): The dependent variable in the current study is the load capacity factor (LCF). The LCF, which gives an overall impression of environmental quality, is calculated by dividing the biocapacity by the ecological footprint, indicating the extent to which demand for natural resources outweighs the regenerating potential of the environment. LCF offers a comprehensive assessment of effectiveness of entire energy system and extends existing indices for environmental quality, primarily GDP based, carbon footprints, and resources depletion indicators.

Energy Transition (ET): The ET variable represents the percentage of renewable energy consumption from total energy consumption. The data has been obtained from the World Development Indicators (CD-ROM, 2023).[1] The transition from fossil fuel consumption to renewable energy sources would enable SSA countries to achieve social, economic, and environmental benefits simultaneously tackling climate change and energy substitution.

The ***Environmental Policy Stringency Index (EPS)*** is a country-specific statistic derived from the OECD (2023),[2] that measures the [illegible] of a nation's environmental policies. Through the inclusion of 14 environmental policy initiatives, which address issues such as climate change and air pollution. Executing an extensive EPS index, SSA countries can ensure a sustainable future, address environmental issues and stimulate economic development.

Financial Development (FD): This parameter represents the level of financial development in a country, specifically measured as the index developed by IMF.[3]

Green Technological Innovation (Total Patents): The number of patents pertaining to green technologies is denoted by this parameter. The data for this variable is collected from OECD statistics.

3.2. Theoretical underpinning

The empirical model in current study is based on the theoretical framework of environmental, economics and sustainability. Thus, we argue that there are four factors that that influence environmental sustainability; ET, INV, FD, and EPS. This rationale suggests that shift from fossils fuels to renewable energy sources enhance environmental sustainability. FD encourages investments in INV thus fostering sustainability. Simultaneously, EPS promotes deployment of INV thus boosting environmental sustainability. Based on these associations we developed an empirical model to evaluate the role of these parameters on environmental sustainability in SSA region. Incorporating these factors enables a thorough assessment of how financial development, environmental policies, technological innovations, and economic dynamics interact to influence the establishment of sustainable development approaches. The basic econometric models are as follows:

[1] World Development Indicators | DataBank (worldbank.org).
[2] OECD Data.
[3] IMF Data.

$$LCF_{it} = \vartheta_0 + \vartheta_1 EPS_{it} + \vartheta_2 INV_{it} + \vartheta_3 ET_{it} + \vartheta_4 FD_{it} + \vartheta_5 EG_{it} + \varepsilon_{it} \quad \text{(i)}$$

$$LCF_{it} = \vartheta_0 + \vartheta_1 EPS_{it} + \vartheta_2 INV_{it} + \vartheta_3 ET_{it} + \vartheta_4 INV^{*}FD_{it} + \vartheta_5 EG_{it} + \varepsilon_{it} \quad \text{(ii)}$$

$$LCF_{it} = \vartheta_0 + \vartheta_1 ET_{it} + \vartheta_2 INV_{it} + \vartheta_3 FD_{it} + \vartheta_4 INV^{*}EPS_{it} + \vartheta_5 EG_{it} + \varepsilon_{it} \quad \text{(iii)}$$

$$LCF_{it} = \vartheta_0 + \vartheta_1 ET_{it} + \vartheta_2 INV_{it} + \vartheta_3 EPS_{it} + \vartheta_4 ET^{*}FD_{it} + \vartheta_5 EG_{it} + \varepsilon_{it} \quad \text{(iv)}$$

INV use advanced exploration techniques to identify environmental technologies. It collects innovations that specifically address environmental management, climate change mitigation, adaptation, and sustainability (Sahoo et al., 2023). Monitoring patents in this field provides valuable information on the technological progress and inventions that contribute to environmental sustainability in the selected economies.

FD Financial development is regarded as a significant determinant of innovative green technology to achieve the environmental sustainability. The FD is an index developed by the International Monetary Fund (IMF). This study incorporates FD as a moderator for GINV and LCF into the empirical model. According to the current research and theoretical framework, this is anticipated that FD has a beneficial influence on green innovations to improve environmental sustainability (Shahid et al., 2024) in SSA countries.

The ***EPS*** measures the level of strictness of environmental rules in each nation. The EPS offers a thorough assessment of the regulatory structure and the extent of dedication towards tackling environmental issues (Li et al., 2023). This variable allows for an evaluation of the impact of strict environmental rules on the countries under investigation. The selection of this variable is due to its empirical significance, theoretical relevance and regional specificity and offer comprehensive policy coverage for SSA region.

The ***ET*** refers to the significant change from traditional energy sources to renewable sources in the manufacture and distribution of energy. Monitoring this transition is essential for comprehending the advancements and effects of adopting renewable energy in the countries under study (Ramzan et al., 2023). Including this variable supports evaluating progress towards a sustainable environment.

Additionally, employing interaction terms of financial development, environmental policy stringency, and key variables such as green technological innovation and energy transition to analyze their impact on LCF is an insightful procedure. This approach supports a thorough assessment of interactions and effects of various factors on LCF, unveiling potential synergies and trade-offs, enabling more effective policy intervention. By analyzing these interactions, policymakers may discern the optimal combination of FD, EPS, and INV deployment to expedite LCF paradigm changes, while also realizing how contextual factors determine the efficacy of these approaches.

Further, these interaction terms provide a more comprehensive investigation into the combined impact of these variables on LCF within the realm of green innovation and energy transition. Each of these factors is systematically selected to provide a thorough comprehension of several aspects that are significant to the study's emphasis on energy transition, financial development, environmental policy stringency, and green technological advancement within the panel Sub-Saharan African economies.

3.3. Long run empirical analysis techniques

The empirical estimation analysis comprises seven steps which are graphically illustrated in Fig. 2 and are described in detail in subsequent sections.

To analyze the long run association among study variables, the study employs dynamic OLS, Fully modified OLS and Canonical Cointegration Regression (CCR) long run estimation techniques. DOLS and FMOLS techniques are selected for their capacity to handle non-stationarity, cointegration, endogeneity, and simultaneity attributes. While CCR accounts for non-stationarity, and allows for testing of cointegration rank and associations. These attributes are pertinent for SSA region dataset due to panel variation, non-stationarity, and interdependent associations selected variables, geographic disparities and non-linear effects. Following equation demonstrates long run cointegration in the study:

$$GHG_{it} = \vartheta_0 + \vartheta_1 EPS_{it} + \vartheta_2 INV_{it} + \vartheta_3 ET_{it} + \vartheta_4 EG_{it} + \vartheta_5 FD^{*}INV_{it} + \sum_{i=-q}^{q}\delta_1 \Delta EPS_{t-i} + \sum_{i=-q}^{q}\delta_2 \Delta INV_{t-i} + \sum_{i=-q}^{q}\delta_3 \Delta FD_{t-i} + \sum_{i=-q}^{q}\delta_4 \Delta EG_{t-i} + \sum_{i=-q}^{q}\delta_5 \Delta INV^{*}FD_{t-i} + \varepsilon_i \ldots.. \quad \text{(v)}$$

$$GHG_{it} = \vartheta_0 + \vartheta_1 FD_{it} + \vartheta_2 ETs_{it} + \vartheta_3 INV_{it} + \vartheta_4 EG_{it} + \vartheta_5 EPS^{*}INV_{it} + \sum_{i=-q}^{q}\delta_1 \Delta FD_{t-i} + \sum_{i=-q}^{q}\delta_2 \Delta INV_{t-i} + \sum_{i=-q}^{q}\delta_3 \Delta ET_{t-i} + \sum_{i=-q}^{q}\delta_4 \Delta EG_{t-i} + \sum_{i=-q}^{q}\delta_5 \Delta INV^{*}EPS_{t-i} + \varepsilon_i \ldots.. \quad \text{(vi)}$$

$$GHG_{it} = \vartheta_0 + \vartheta_1 EPS_{it} + \vartheta_2 ETs_{it} + \vartheta_3 INV_{it} + \vartheta_4 EG_{it} + \vartheta_5 ET^{*}FD_{it} + \sum_{i=-q}^{q}\delta_1 \Delta EPS_{t-i} + \sum_{i=-q}^{q}\delta_2 \Delta INV_{t-i} + \sum_{i=-q}^{q}\delta_3 \Delta ET_{t-i} + \sum_{i=-q}^{q}\delta_4 \Delta EG_{t-i} + \sum_{i=-q}^{q}\delta_5 \Delta ET^{*}FD_{t-i} + \varepsilon_i \ldots.. \quad \text{(vii)}$$

Further, this research expands its investigation of the mechanisms governing LCF by utilizing the MMQR approach, as suggested by Machado and Silva (2019), with fixed effects. The significant aim is to empirically analyze the dynamic relationships between energy transition (ET), environmental policy stringency (EPS), green technology innovation (GT), and financial development. Furthermore, it seeks to investigate the interactive association between green innovation,

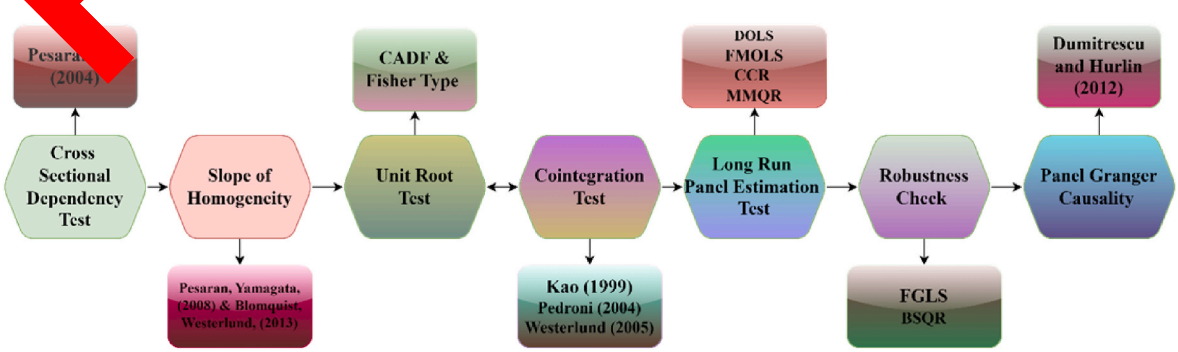

Fig. 2. Empirical estimation analysis roadmap.

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financial development, and environmental policy strictness in the economies of Sub-Saharan Africa. By employing this method, we intend to enhance our comprehension of the factors influencing sustainable finance. Conventional panel quantile regression methods can identify the existence of unobserved variability across the panel, but they are unable to identify outliers. According to Galvao Jr (2011) and other researchers, this strategy necessarily considers the effect of the covariance of conditional heterogeneous LCF determinants. It emphasizes the particular consequence that affects the overall result, rather than simply changing the methodologies. This technique can effectively address the possibility of explanatory variables, such as endogenous features. This method is applicable when the impacts of individual factors are deeply integrated into the panel data framework. The MMQR method provides reliable estimates in the case of a nonlinear equation for obtaining non-crossing estimates in quantile regression. The study's framework is derived from the work of Pata and Samour (2023), and it is presented as follows:

$$\boldsymbol{Y}_{it} = \vartheta_i + \boldsymbol{W}_{it}\boldsymbol{\delta} + (\gamma_i + \boldsymbol{X}_{it}\boldsymbol{\nu})\boldsymbol{U}_{it} \tag{ix}$$

ϑ,γ, δ, and ν indicates the vector of parameters whereas, probability expression is $p(\gamma i*Xit*\tau > 0)$. The fixed effect of individual i are values with (ϑ *i, γi),* where i = 1,2,3, ….n. X matrix represents the vector *K* composed of specified components *W*, which are distinct transformations. The ith element in X is explicitly specified as follows:

$$\mathrm{Xl} = \mathrm{Xl}(\mathrm{W})\ ,\ \ \mathrm{l} = 1, \ldots, \mathrm{k}$$

The primary model posits that the cognitive ability factor is uniformly distributed and exhibits consistent attributes for any given individual at a particular moment. Similarly, the uit factor exhibits a symmetrical distribution across all persons and time points. It is also standardized to comply with the set parameters by Machado and Silva (2019).

$$\mathrm{QY}\ (\tau\,|\boldsymbol{W}_{it}) = \vartheta_i + \gamma_i \boldsymbol{q}(\boldsymbol{\tau})) + (\boldsymbol{W}_{it}\delta + \boldsymbol{X}_{it}\boldsymbol{\nu}\boldsymbol{q}(\boldsymbol{\tau})$$

The array of vectors Xit depicts the explanatory variables, encompassing environmental policy stringency, economic growth, financial development, and green technical innovations. Conversely, [illegible] (τ/Xit) reflects the explained variable load capacity factor [illegible] the ab[illegible] (x).

To validate the accuracy of outcomes from selected techniques we executed robustness tests with FGLS and QR. FGLS deals with heteroskedasticity and serial correlation, whereas QR offers distributional intuitions and handles non-normality. These techniques are chosen due to their aptitude to handle nonstandard errors, resilience to outliers and non-normal distributions and adaptability in depicting intricate associations among variables. Both selected techniques validated the main findings. Further, GMM and system GMM indicated similar coefficient signs and indications but exhibited lower preciseness. However, IV produced contradictory results potentially due to inadequate instrumentation.

4. Empirical results

This section explains the results obtained from extensive econometric techniques. Table 1 illustrates the descriptive statistics of the parameters.

The results indicates that the ET has the low mean value of 1.845337 with standard deviation of 1.321683, as determined by the major composite analysis that incorporates standardized variables. On the other hand, environmental quality has the greatest mean value of 44.66635 with standard deviation 26.1852. The minimum value of 1.27958 and maximum value of 100.0.

To ascertain the absence of collinearity among the parameters of the model, this study employs correlation analysis. The results are summarized in Table 2. These results indicates that all the values are less then threshold 0.8. this suggests there is no issue of multicollinearity as the highest correlation value is 0.6841 between EPS and LCF.

This study also employs the VIF test to check the issue of multicollinearity among study variables. The VIF quantifies the relationship between the partial correlation coefficient and the rise in variance and covariance of the estimates for sets of variables. As the partial correlation coefficients approach a value of one, the VIF (Variance Inflation Factor) increases to infinity. Consequently, the variance and covariance also increase without limit, causing the estimates to become inconsistent and unbiased.

The econometric models used in this research effectively mitigate collinearity and ensure independence within the cross sections by employing panel estimation techniques. The results in Table 3 indicate that following the transformation of the variables, the Variance Inflation Factor (VIF) approaches unity. This shows that the issue of collinearity between the variables is no longer a concern for subsequent econometric estimation.

The CD test results in Table 4 indicates the existence of CD in panel data, thus rejecting the null hypothesis at a significance level of 1%. The approach for detecting cross-sectional dependency reveals a significant presence of interdependence. This suggests that the time series exhibits contingency effects, meaning that disruptions and shifts in one country can affect other cross sections due to the interconnected nature of countries.

Table 5 displays the findings of the slope homogeneity analysis. The empirical values of delta and adjusted delta of both tests Pesaran and Yamagata (2008) and Blomquist and Westerlund (2013) suggest the presence of a heterogeneous slope.

The study utilizes second-generation unit root tests, primarily Fisher-type and CADF unit root tests, to examine the presence of the unit root problem among the variables. The unit root test findings, reported in Table 6, indicate that all variables exhibit unit root issues at the level. However, all the variables become stable at I(1).

After confirming the stationarity of variables, this study performed the Kao, Pedroni and Westerlund cointegration tests in order to check the long run association among the variables. Thes results are summarized in Table 7. The p-value of three tests is significant at 5% and 1% level indicating that there is long run association among the study variables. This results thus confirm that further long run econometric estimations techniques can be employed.

Furthermore, the outcomes of long-run elasticities of several parameters are displayed in Table 8, derived from DOLS, FMOLS, and CCR estimations. The results indicate that 1% increase in enactment of strict environmental regulation contributes to improve environmental sustainability by .4970042 in DOLS, .5196633 in FMOLS and .5198218 in

Table 1
Descriptive statistics.

Variable	Mean	Std. Dev.	Min	Max
LCF	44.66635	26.1852	1.27958	100
EPS	16.85606	25.43699	0.445986	126.7
INV	7.432247	3.488885	1.63522	22.2082
ET	1.845337	1.321683	−3.901083	5.639386
GDP	3.657915	3.612745	0.000787	17.7716
FD	23.30012	1.359698	20.37	26.96

Table 2
Correlation matrix.

	lcf	eps	inv	et	gd	fd
LCF	1.0000					
EPS	0.6841	1.0000				
INV	0.4615	0.4117	1.0000			
ET	0.0714	−0.1471	−0.1826	1.0000		
GDP	−0.1694	−0.1257	0.0588	0.0694	1.0000	
FD	0.0991	0.1639	−0.4171	0.0183	0.0342	1.0000

Table 3
VIF test.

Variable	VIF	1/VIF
EPS	1.79	0.557886
INV	1.51	0.662432
ET	1.48	0.674244
GDP	1.06	0.943435
FD	1.05	0.953755
Mean VIF	1.38	

Table 4
Average correlation coefficients & Pesaran et al. (2004) CD test.

Variable	CD-test	p-value	corr	abs(corr)
LCF	69.56	0.000	0.806	0.808
EPS	22.11	0.000	0.255	0.520
INV	7.27	0.000	0.079	0.389
ET	1.35	0.000	0.013	0.399
GDP	19.49	0.000	0.225	0.370
FD	78.93	0.000	0.907	0.907

Notes: Under the null hypothesis of cross-section independence CD ∼ N(0,1).

Table 5
Slope of homogeneity.

Pesaran, Yamagata. 2008		
	Delta	p-value
Δ	11.566	0.000
Δ~Adjusted	14.486	0.000
Blomquist, Westerlund. 2013		
Δ	9.700	0.000
Δ~Adjusted	12.148	0.0[illegible]

Table 6
Unit root test.

	Fisher Type		CA[illegible]	
Variables	I(0)	I(I)	[illegible]	[illegible]
LCF	2.7478	−28.0511***	0.79[illegible]	−4.304***
EPS	0.8706	−16.8826**[illegible]	0.151	−1.257***
INV	−1.4611	−18.371[illegible]	−0.809	[illegible]4.637***
ET	−4.1354*	−18.3[illegible]	2.070	[illegible]4.599***
GDP	−2.0019	−17.6[illegible]42***	[illegible]0.248	−4.447***
FD	1.1948	[illegible]472***	[illegible]	−1.022***

Table 7
Test for cointegration.

	Statistic	p-value
Kao test		
Modified [illegible]-Fuller [illegible]	−21.1956	0.0000
Dickey-Fuller [illegible]	−26.4147	0.0000
Augmented Dic[illegible]ler t	−16.6575	0.0000
Unadjusted modifi[illegible]key-Fuller t	−33.3115	0.0000
Unadjusted Dickey-Fu[illegible]	−28.1481	0.0000
Pedroni Test		
Modified Phillips-Perron t	2.2168	0.0133
Phillips-Perron t	−23.9509	0.0000
Augmented Dickey-Fuller t	−17.2361	0.0000
Westerlund test		
Variance ratio	50.0065	0.0000

CCR. Similarly, 1% increase green innovation, increase the environmental quality by 3.012985, 2.779133, 2.796868 as per DOLS, FMOLS and CCR, respectively. While energy transition also helps to foster environmental sustainability by 2.229982, 2.503006 and 2.54956 as per results derived from DOLS, FMOLS and CCR, respectively. The substantial influence of INV and ET highlights the necessity of thorough R&D, and regulatory reforms, to attain economic sustainability. These efforts are crucial for achieving long-term structural transformations. These study findings are aligned with the study of Liao et al. (2023). Their study examined the renewable energy transition and industrialization on environmental sustainability in OECD countries. They suggested that presence of industrialization and GINV has a significant effect, indicating that the implementation of GINV policy counteracts the negative consequences of manufacturing processes and adopting green innovative technologies and shifting to green energy mitigates environmental degradation. Enhancing the energy transition in conjunction with adopting green initiatives would significantly reduce several levels of environmental degradation.

Moreover, FD also has a significant [illegible] positive impact on environmental quality. The results of DOLS, FMOLS and CCR indicate that 1% increase in FD increases environmental sustainability in SSA countries by 3.305418, 2.926219, 2.9[illegible]428, respectively. These results indicates that FD plays a crucial role in ensuring the economic progress of a country and creates significant development opportunities. The advancement of sustainable finance leads to an increase in environmental sustainability. Environmentally focused FD policies have the potential to lower pollution levels in Sub-Saharan African (SSA) countries. These findings are consistent with the discoveries of (Jiakui et al., 2023). They suggest that FD has a positive impact on reducing pollution by promoting green practices and promoting environmental quality. Significant financial growth that encourages environmental concerns leads to a reduction in pollution.

The interaction of FD*INV possess a positive impact on environmental sustainability by .1068527, .0994413, and .1002256 in DOLS, FMOLS, and CCR outcomes. This indicates that financial reform and development [illegible] encourage foreign direct investment and increase investments in green technology, leading to accelerated economic growth and impacting sustainability. Another argument is that financial development enables developing nations to access and utilize new technology, supporting them in adopting cleaner and more environmentally friendly industrial methods. This, consequently, contributes to the overall improvement of the global environment and enhances the sustainability of economy. Furthermore, the results of interaction of EPS*INV also contributes to improve the environmental sustainability by .0634727 in DOLS, .0650288 in FMOLS and .0652332 in CCR. This indicates that EPS exerts a beneficial influence on both the trajectory and pace of technological advancement. Businesses pursuing to fulfill advanced technological requirements and adhering to government-mandated environmental regulations will prioritize the development of environmentally friendly innovations. By doing so, they gain a competitive edge and also contribute to the green technology innovation process of less advanced businesses by offering their innovative products. These findings are supported with the study of Hasan and Du (2023). Their research investigates the impact of green financial development, technical innovation, and environmental legislation on China's efforts towards achieving sustainable development. They demonstrated that understanding the impact of environmental regulations on the relationship between economic stability and skewed technological innovation is essential. Green financial innovation and green technical innovation are crucial strategies for attaining environmental sustainability.

Moreover, the current study employed FGLS (feasible generalized least square) technique to check the robustness of the DOLS, FMOLS and CCR. The EPS, INV, ET and FD contribute to increase environmental sustainability. While EG has negative impact on environmental quality due to increase of human activities. The results indicates that the interaction of FD*INV demonstrates a strong association with LCF, leading to improved efficiency in business investment in green innovative technologies and ultimately contributing to environmental sustainability. Furthermore, coefficients of EPS*INV also demonstrate positive association with environmental sustainability which indicates

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Table 8
Panel cointegration tests.

Variables	DOLS	FMOLS	CCR	FGLS
EPS	.4970042*** (.1084163)	.5196633*** (.1074213)	.5198218*** (.1078433)	0.525032*** (.0360836)
INV	3.012985*** (.9413798)	2.779133*** (.8801496)	2.796868*** (.9070021)	2.589504*** (.2938176)
ET	2.229982*** (1.865646)	2.503006*** (1.747655)	2.54956*** (1.807377)	(1.444667*** (.5875728)
GDP	−.9580743*** (.7085972)	−.7649553*** (.6856059)	−.7795198*** (.6918939)	−1.051064*** (.2292063)
FD	3.305418*** (2.068246)	2.926219*** (2.017937)	2.940428*** (2.046631)	3.091294*** (.6764888)
FD*INV	.1068527** (.039624)	.0994413** (.0358169)	.1002256*** (.0368741)	.1127011*** (.013201)
EPS*INV	.0634727** (.0089505)	.0650288** (.0089256)	.0652332*** (.0090281)	.0622746*** (.0027911)
ET*FD	.1676119** (.2108559)	.1037492*** (.167089)	.0980643*** (.1752124)	.1504492*** (.0529005)
_cons	−64.02812** (51.735)	−55.21745** (50.18074)	−55.72637*** (51.04625)	−54.6407*** (16.79934)

that the advancement of environmentally friendly technologies is increasingly becoming a fundamental aspect of environmental policy and the most promising solution for achieving sustainable development and an environment free from pollution. Green technologies have been recognized as a key objective of the UN SDGs agenda. This is because global warming is a worldwide issue, and there is a growing need for international environmental policies that promote collaboration in the advancement of green technologies to combat global climate change and pollution.

Moreover, coefficients of ET*FD demonstrate a positive relationship with LCF. This demonstrates that 1% change in ET*Fd increases LCF by .1676119 in DOLS, .1037492 in FMOLS and .0980643 in CCR. This posits that FD is vital in fostering transition from fossils fuels to renewable energy sources by enabling investment in renewable technologies, innovation and infrastructure. Increased availability to financial sources enables efficient allocation of funds for renewable projects, such as solar, wind farms and hence lowering reliance on fossil fuels. Advanced financial instruments, i.e, green investments, green stocks and green bonds capture the interest of institutional and typical inv[illegible] aligning financial benefits with environmental objectives. FD str[illegible]lines the transition from conventional energy sources to renewable [illegible]ergy sources, simultaneously, driving economic growth, employme[illegible] and adaptation to environmental changes, thus contri[illegible] a more sustainable future.

4.1. MMQR analysis

Furthermore, the method of m[illegible]f quantile [illegible]ssion is employed to examine the associati[illegible] of LC[illegible] the SSA nations across different quantiles. The results [illegible]MQR are su[illegible]ized in Table 9. The results demonstrate that EP[illegible]V, ET, and FD ha[illegible] positive and statistically significant im[illegible] on L[illegible] across all quantiles. Based on Table 9, EPS has incre[illegible] eff[illegible]n LCF ranging from .4647729 to .5888763 in all quantiles [illegible]8). This[illegible]cates that strict environmental regulati[illegible] the f[illegible]ent[illegible]illars of environmental sustainability a[illegible] sol[illegible]n for m[illegible]g environmental degradation. These fi[illegible]s are [illegible]gned with the study of (Wolde-Rufael and Mulat-Wel[illegible]ske[illegible] indicated that the objective of strict regulations is to increase the price of pollut[illegible] environmental services, to influence the behavior of [illegible] industri[illegible]d consumers toward the adoption of more environm[illegible]ly friendly [illegible]ucts. This is achieved by the implementation [illegible] regul[illegible]s that im[illegible] limitations on polluting agents, thus increa[illegible] the finan[illegible]urd[illegible]ssociated with polluting activities and red[illegible]g thei[illegible]nticem[illegible]milarly, INV, ET and FD are positively ass[illegible]ed wi[illegible]e environmental sustainability. The INV has 2.68909[illegible] 2.6[illegible].589653[illegible]d 2.483986 impact on LCF from quantile [illegible]o 0.8. T[illegible] has 1[illegible]124, 1.207928, 1.444041 and 1.887053 f[illegible]-0.2 to Q-0.[illegible]

The study [illegible] that EG has a[illegible]gative impact on environmental sustainability by −1.2[illegible]7, −1.14440, −1.051311 and −.8766357 in all qua[illegible] from 0.2 to [illegible] This is due to increased economic growth lead[illegible]er levels of pro[illegible]tion and consumption, resulting in more w[illegible], pollution, and strain on natural resources. The environment, co[illegible]mption of [illegible]urces, and economic activity are intricately interco[illegible]ted. They [illegible]ntend that although the utilization of resources, parti[illegible] en[illegible] sources, brings about immediate economic advantages, its [illegible]ental effects on the environment may become apparent [illegible] time.

[illegible]eraction of FD*INV demonstrates a positive and significant impact on LCF ranging from .1074288, .1003036, .0909601 and .0725624 from quantile 0.2 to 0.8. This suggests that FD promotes technical innovations, which are a significant source of effectiveness, by allowing the mobilization of money and sharing of uncertainties. Hence, strengthening the financial system through a more open capital account can potentially promote the effective utilization of technology and thus impact financial development and environmental deterioration alleviating infrastructural deficit in SSA region. This indicates that FD and INV policies should emphasize infrastructural advancement, transfer of technologies, and capacity development. These findings indicate that tailored policy interventions can offer significant environmental perks, regardless of infrastructural and innovative technological challenges.

Financial support for the process of restructuring and improving the economy and promoting the growth of sustainable businesses through the use innovative technologies. Sustainable environmental growth requires the efficient utilization of technological developments that benefit society and satisfy the requirements of present and future

Table 9
MMQR analysis.

	Quantiles			
Variables	0.2	0.4	0.6	0.8
EPS	.4647729*** (.0335059)	.4908661*** (.0353426	.5249415*** (.0443431)	.5888763*** (.0706709)
INV	2.689096*** (.2341789)	2.645971*** (.2452278	2.589653*** (.306908)	2.483986*** (.4909445)
ET	1.027124*** (.4403676)	1.207928*** (.4618151	1.444041*** (.5782815)	1.887053*** (.9243164)
GDP	−1.215697*** (.1872234)	−1.14440*** (.1963129	−1.051311*** (.2458048)	−.8766357*** (.3929301)
FD	3.169504*** (.5147719)	3.13564** (.5388255	3.091411*** (.6742542)	3.00843*** (1.07880)
FD*INV	.1074288*** (.0091165)	.1003036*** (.0096791	.0909601*** (.0120812)	.0725624*** (.0192022)
EPS*INV	.0575812*** (.0032348)	.060511*** (.0031624	.0624464*** (.0036965)	.0664835*** (.0056498)
ET*FD	.1850782*** (.0404342)	.1703305*** (.0421256)	.1494445*** (.0524229)	.1132478*** (.0810303)
_cons	−70.21671*** (12.73084)	−63.472*** (13.34842	−54.66405*** (16.71683)	−38.1379*** (26.71538)

Note: ****, **, and * denote 1, 5, and 10% significance levels, respectively. Standard errors are in the parenthesis.*

generations. The significance of FD is crucial to support technical development in attaining the sustainability of the environment. To attain environmental sustainability targets, businesses must cultivate their understanding of innovation, internal advancement of technology, and competency to effectively utilize financial resources to make high-quality FD. These findings aligned the conclusion of Rajguru & Kautish (2023) which argue that FD promotes technical innovations which are a significant driver of productivity, by streamlining the mobilization of funds and tackling uncertainties.

Furthermore, the interaction of EPS*INV also demonstrates positive association with LCF. The outcomes demonstrate .0575812, .060511, .0624464 and .0664835 from quantile 0.20 to 0.80. This indicates EPS has effectively curtailed pollution triggered by businesses and have contributed a crucial role in protecting the environment. EPS can serve as an interface connecting sustainable innovation with the adaptive enhancement of manufacturing businesses. This indicates that a 1% increase in EPS*INV, increases environmental sustainability by 5% over a decade. These implications need policy intervention, particularly in the underdeveloped infrastructure of SSA. These practical assessments indicate that certain stringent strategies could offer substantial environmental benefits. This posits that environmental regulations and incentives frequently encourage green technologies and practices for instance renewable energy, energy efficiency, and emission reduction. Moreover, a supportive, feasible, and rapidly implementable policy framework can encourage and facilitate, technological advancement, a shift in public and industrial conduct towards sustainable economy and environment. EPS might raise adherence costs, however, the potential benefits may outweigh these expenses. Numerous studies have demonstrated that implementation of environmental rules and regulations strictly promote economic efficiency and encourage green innovative technologies and practices. Consequently, EPS are more likely to encourage the adoption of green innovative technologies and enhance environmental quality. This implies that EPS with green technological adoption contribute to improve environmental sustainability in SSA countries. EPS frequently stimulates development of efficient and innovative technologies that lesson environmental degradation. However, it depends on various factors such as, the nature of innovation, its adoption and implementation, and its regulatory [illegible] the impact of GINV on environmental sustainability [illegible] SSA region. For instance, the implementation of renewable energy technologies and emission capture and [illegible] technologies [illegible] reduce degradation during energy production. Similarly, GINV may stimulate efficient use of energy and lower associated emission in manufacturing, transportation and construction industries.

The Porter hypothesis is a commonly acknowledged framework for examining the association of environmental regulations and the development of green technological innovation. It is of the opinion that environmental regulations has a positive impact on innovation by providing incentives for the advancement of [illegible] technology, the creation of green technology, and the dissemination of emissions reduction technologies, ultimately [illegible] an improvement in environmental quality. In [illegible] region, FD facilitates green finance accessibility, fostering innovation and businesses activities. EPS stimulates adoption of innovative technologies, encouraging sustainable projects. The Porter's hypothesis claims that FD derives green innovation, consequently improves environmental quality. However, the distinct challenges of climate change, biodiversity depletion, resources reliance and inadequate financial source access required tailored regulatory strategies in SSA region. The study findings supports relevance and applicability of the Porter's hypothesis in SSAregion by analyzing institutional factors, inadequate regulatory structures, and management constraints. By synthesizing these theoretical framework, the study offer significant perspectives into the intricate associations FD, EPS, INV and ET in SSA region.

These findings are in line with the study of Ouyang et al. (2020). They posit that regulatory bodies in developing countries can effectively utilize green innovative technologies by implementing strict environmental regulations and organized initiatives that reveal facts about businesses' environmental sustainability. Thus, based on study findings it can be argued that in SSA region, EPS is crucial for stimulating GINV and addressing significant environmental challenges. Economic intuitions show that EPS could promote the adoption of INV by encouraging industries to allocate resources for R&D to adhere to regulations, consequently catalyzing innovations in green technologies. Adoption of INV fosters business viability and growth thus strengthening their potential to compete in internal markets. Thus, these findings support the existence of Porter's hypothesis in the SSAregion.

Furthermore, the coefficients of the interaction term ET*FD are positively associated with environmental sustainability. This reveals that a 1% increase in ET*FD increases environmental quality by .1850782, .1703305, .1494445, and [illegible]2478 from quantile 20th to 80th. This indicates that in SSA countries FD plays a [illegible] moderating role in driving ET for a sustainable environment. The [illegible] to expedite sustainable development robustly and substantially requires transition from conventional energy sources to renewable energy sources. According to IEA report, solar and wind energy is anticipated to increase eight times in SSA region. Further, IRENA report estimated that to reach temperature till1[illegible] RE shares [illegible] increase up to 90% in 2050. Amidst the constriction of economic situation in 2022, including interest rates, supply chain disruptions and inflations, ET appears as the catalyst for sustainable development. FD plays a crucial role in stimulating ET, encouraging SSA region to utilize renewable energy sources, minimize reliance on fossil fuels and achieve environmental sustainability. Effective FD policies will determine the potential of SSA region for ET and whether the energy structure of SSA region is adaptable, accessible, and sustainable. [illegible] government, policymakers and businesses possess the potential to collaborate to initiate ET, contributing as a catalyst for sustainable development.

Therefore, environmental challenges such as deserts, and degradation require appropriate green innovative technologies in SSA region. However, a distinct environment poses challenges. Inadequate regulatory structures restrict the effectiveness of EPS. Increasing regulatory costs might discourage industries and businesses from pursuing R&D activities. Insufficient infrastructure restricts the adoption and implementation of INV. To overcome these obstacles developing policy structures, strengthening regulatory bodies, and redesigning policies is crucial. Thus, appropriate resources allocation and incentives i.e., tax rebates and subsidies could encourage GINV. Joint initiatives, such as the African Union's Green Economy Strategy could stimulate the dissemination of knowledge and collaborations. Furthermore, in SSA region, FD is crucial for fostering implementation and development of GINV. Access to finance encourages businesses to invest in R&D to promote GINV. Advanced and innovative financial instruments assist in managing risk associated with GINV. FD stimulates innovations, boost development and competition in GINV in SSA region. Easy access to finances, mitigation of risk, dynamic competition encourage sustainability. Thus, Tailored perks, regulatory structures, and capacity development tackle challenges. Fig. 3 demonstrates that the trends of environmental quality contributed by independent variables across different quantiles.

4.2. Robustness check of MMQR approach

The present study used BSQR approach to validate the results of MMQR analysis. The results indicate that EPS, INV, ET, and FD are positively associated with LCF. This suggests that enactment of strict environmental regulations, green innovation, shift from conventional energy sources to renewable sources and developed financial structure contributes to increase the environmental sustainability. Moreover, the interaction of FD*INV and EPS*INV also contributes positively by increasing environmental quality in SSA countries. The results in Table 10 validate the results of MMQR approach.

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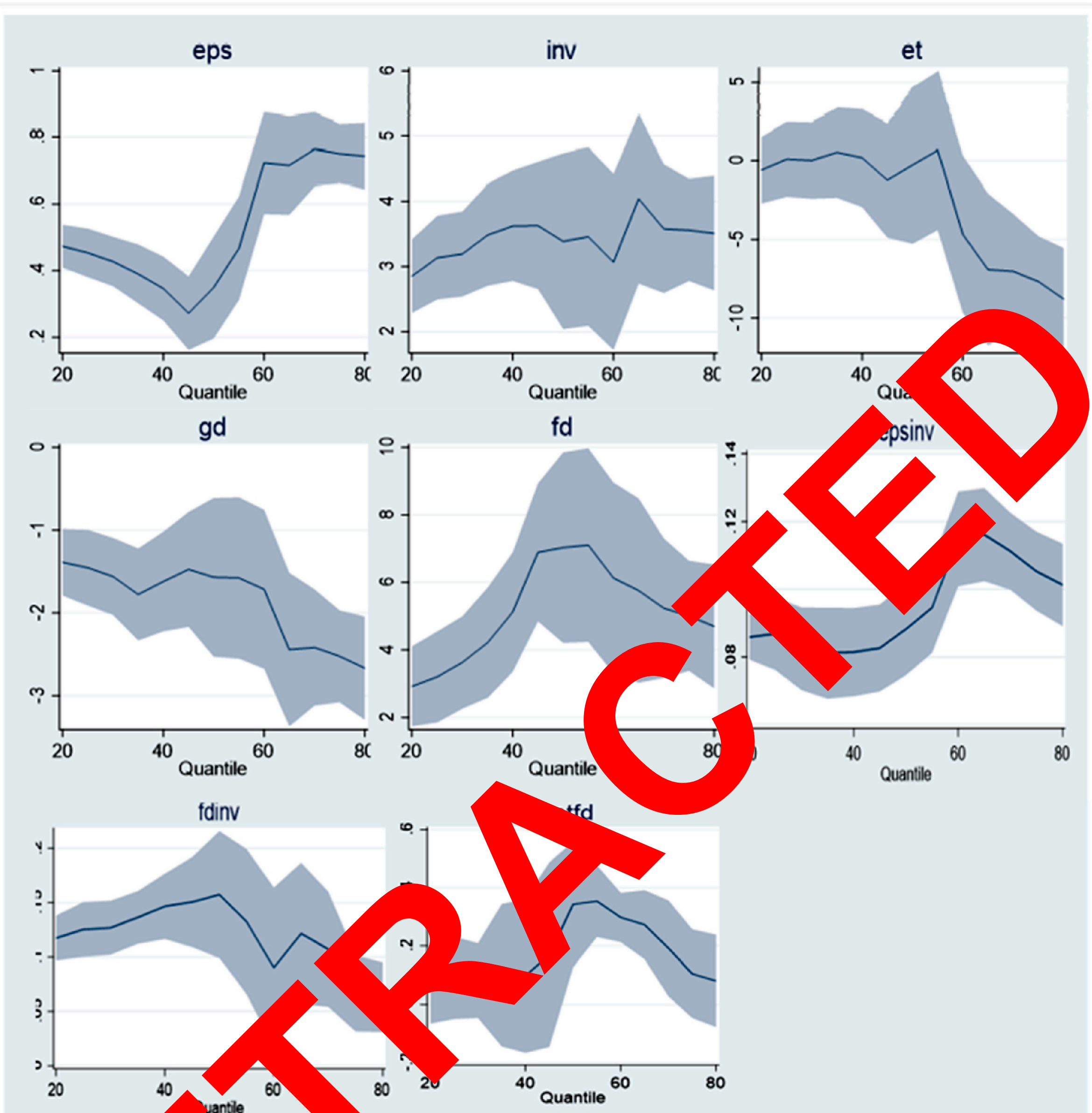

Fig. [illegible] Graphical illustration of quantile regression outcomes of selected series of variables.

Table 10
Non parametric [illegible]

	[illegible]ntiles			
Variables	[illegible]	0.4	0.6	0.8
EPS	.5094045*** (.0199101)	.4026024*** (.0966977)	.6099782*** (.1623437)	.7893012*** (.1633249)
INV	[illegible].663685*** (.1929266)	3.411686*** (.4177648)	2.730261*** (.6193368)	2.278276*** (.9856475)
ET	[illegible]79667*** (.3699108)	1.503832*** (.5429766)	4.018911*** (.4939199)	3.340468*** (2.373099)
GDP	[illegible]1.197691*** (.2347916)	−1.455222*** (.2661365)	−1.482682*** (.3813493)	−1.958416*** (.3033321)
FD	2.204359*** (.576105)	4.096367*** (.8539303)	5.871482*** (.5945041)	4.251384*** (1.550514)
FD*INV	.1135626*** (.0063064)	.1235483*** (.008317)	.0693161*** (.0234084)	.0317756*** (.0353027)
EPS*INV	0.0608171*** (.0019543)	.0589806*** (.0034438)	.0643289*** (.0066928)	.0918873*** (.0092328)
ET*FD	.1122424*** (.0701726)	.977464*** (.0937459)	.2975681*** (.0545552)	.803720*** (.1028684)
_cons	−48.52806*** (13.38908)	−89.63463*** (18.83881)	−123.977*** (14.40473)	−69.44443*** (42.69009)

Note: ****, **, and * denote 1, 5, and 10% significance levels, respectively. Standard errors are in the parenthesis.*

These robust outcomes affirm the reliability, correctness of our findings, supporting the crucial role of FD and EPSfor fostering INV in SSAeconomies.

4.3. Panel causality test analysis

Finally, we analyze the causal relationship between the explanatory variables and the explained variable. The MMQR method reveals the

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connections between variables at different quantiles, but it does not provide a causal explanation for these associations. To develop effective strategies, it is essential to understand the paths of these interactions. Considering the presence of the CSD among the variables, we can utilize conventional causality approaches, such as the causality test developed by Dumitrescu and Hurlin (2012) to analyze causality among study variables. This test considers the presence of different characteristics in panel data and establishes a causal relationship by conducting separate regressions for each set of data, as compared to previous tests that examine panel causality as a whole. Further, the W-bar is used to calculate average statistics, whereas Z-bar statistics, which represent a typical distribution of normality, are used to evaluate the significance of the causality. The outcome of this investigation is summarized in Table 11.

Results demonstrate that the independent variables substantially influence LCF, suggesting the presence of a bidirectional causal relationship among them in SSA countries, except GDP. GDP exhibits a unidirectional correlation with LCF.

5. Conclusion and policy recommendations

Enhancing environmental quality is a crucial factor in the sustainable development goals agenda launched by the UN in 2015. Following this schedule, nearly all nations are increasing their efforts to enhance the quality of the environment. Empirical research has emphasized that energy transition and technological innovation play a significant role in improving environmental quality. Within the context of the SSA, there is limited discussion and considerable contention over the association between energy transition, technological innovation and environmental quality. This research aims to reassess the relationship between multiple factors that influence environmental quality in Sub-Saharan Af[illegible] (SSA). For this undertaking, the current study employs a panel d[illegible] comprising 29 nations in Sub-Saharan Africa (SSA) from 1990 to 2[illegible] Previous research conducted in SSA has ignored the moderating role [illegible] financial development between energy transition and [illegible]ronmenta[illegible] quality as well as financial development and the [illegible]gen[illegible] of environmental policies between technological inn[illegible]on an[illegible] environmental quality in the selected countries. Fu[illegible]more[illegible] determines the direction of causality bet[illegible]n fa[illegible]at influence environmental quality.

This research reveals that parame[illegible]S, INV, ET, a[illegible]D have a positive association with environm[illegible]al s[illegible]ability, indicating that they contribute to improving e[illegible]ironmental [illegible]ty in SSA countries. This indicates that there is [illegible]ositive associati[illegible]etween improved environmental condition[illegible]n Sub[illegible]haran Afric[illegible] (SSA) and the enhancement of the depth, efficacy, and accessibility of the financial sector. Furthermore, the findings indicate that financial development has a positive and substantial moderating impact on the relationship between green innovative technologies and load capacity factor. This finding demonstrates that providing financial support and implementing environmental rules to support innovations in technologies leads to enhanced environmental quality in Sub-Saharan Africa. Furthermore, the findings suggest that the level of strictness in environmental policies has a moderating impact on the relationship between green technological development and load capacity factor. Hence, countries in Sub-Saharan Africa (SSA) that exhibit greater levels of technological innovation and energy transition generally experience an improvement in environmental quality. Finally, the results in[illegible]hat the relationship between explanatory variables and en[illegible]men[illegible]ustainability is bidirectional, except for GDP. This re[illegible]ch indicate[illegible] explanatory variables and environmental sustainabi[illegible]use each o[illegible] in SSA. The findings suggest that the long-te[illegible]object[illegible]f enviro[illegible]ental protection should involve implem[illegible]ng a robu[illegible]rat[illegible] that includes strengthening regulations [illegible]u et al[illegible]022), su[illegible]ting clean energy sources, transitioning tow[illegible] a [illegible]n economy, and enhancing efficiency in energy con[illegible]ptio[illegible]

Furthermore, [illegible]y findings [illegible]hasi[illegible] that EPS has significance for redirecting [illegible] developmen[illegible]sustainability in Africa. Effective regulato[illegible]struc[illegible] offer distinc[illegible] specifications for the adoption of green innovative techn[illegible]ies, requiring adherence to environmental sustai[illegible] standards a[illegible]xtensive effect investigations to assess the[illegible]mpact on the environment. Effective pollution control regulations o[illegible]ossil fuels a[illegible] adequate financing stimulate the transition to re[illegible]able energ[illegible]ources, while policies enabling tax rebates, incen[illegible] or ade[illegible]e funding for green innovative technologies promote [illegible]tal sustainability. Further, strong oversight and regulatory framework are crucial to ensure commitment to these stan[illegible] African countries can develop robust financial sectors and sus[illegible] policies that stimulate energy transition, and encourage the adoption of green innovative technologies for promoting environmental sustainability through the implementation of strict environmental policies. Investment in innovative renewable technologies, for instance, smart grids, energy storage, wind, and solar-can enhance renewable energy accessibility and minimize reliability on fossil fuels. By implementing an optimal regulation system and encouraging technological innovation, SSA can exploit its sustainable potential and significantly promote renewable energy consumption.

In light of the study results, the following policy implications have been drawn. Africa is at a critical stage of its economic development, with energy being crucial to its future growth. Considering Africa's extensive renewable resources, the continent has a substantial deficit in energy accessibility. Inadequate funding increases this issue since the region has received just 2% of global green energy investments over the last two decades. Governments of SSA countries need to establish tax benefits for investing in green energy sources. Develop green banks and professional financial institutions committed to green infrastructure projects for instance; urban forestry programs and green roofs. Promoting microfinance schemes for modest renewable energy businesses. Sufficient funding and effective policies are essential for tackling the challenge of an energy deficit by optimizing renewable sources and adopting green innovative technologies. Significant strategies include designing effective regulatory structures to encourage investment, fostering the development of public-private collaborations for resource distributions, and development innovative financing instruments, for instance, green bonds, and green stocks to alleviate financial constraints. Policymakers need to design and execute regulations that foster optimal development of land and environmental sustainability. Establish carbon pricing strategies to encourage sustainable growth. Further, developments in infrastructure, particularly, grid elongation are crucial for the transition to renewable energy sources. Effective renewable energy policies could support remote regions through customized investment

Table 11
Dumitrescu and [illegible]2) Caus[illegible]

Causality [illegible]ions	[illegible]stat	Z-[illegible]tat	Direction	Probability
LCF ≈ EPS	[illegible]	7.9354	**LCF⇄EPS**	0.0000
EPS ≈ LCF	2.2202***	3.2108		0.0013
LCF ≈ INV	1.7321***	1.7561	**LCF⇄INV**	0.0791
INV ≈ LCF	[illegible]91***	3.4216		0.0006
LCF ≈ ET	[illegible]728***	5.1556	**LCF⇄ET**	0.0000
ET ≈ LCF	1.9175***	2.3086		0.0210
LCF ≈ GDP	0.7603**	−1.1400	**LCF⇒GDP**	0.0254
GDP ≈ LCF	1.6319	1.4576		0.1450
LCF ≈ FD	13.6927**	37.4004	**LCF⇄FD**	0.0000
FD ≈ LCF	1.8939**	2.2381		0.0252
LCF ≈ FD*INV	1.8062**	1.9769	**LCF⇄FDINV**	0.0481
FD*INV ≈ LCF	2.3874**	3.7088		0.0002
LCF ≈ EPS*INV	2.487**	4.0059	**LCF⇄EPSINV**	0.0001
EPS*INV ≈ LCF	1.8989**	2.253		0.0243
LCF ≈ ET*FD	3.2435***	8.5431	**LCF⇄ETFD**	0.0000
ET*FD ≈ LCF	1.8356**	3.1819		0.0015

Note: **** and ** represent the rejection of the null hypothesis at the 1% and 5% significance levels, respectively.*

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initiatives. Further, investing in regional capacity development and encouraging regional collaborations in energy projects could enhance robustness and sustainability. By emphasizing these domains, African countries could potentially foster an adequate and sustainable future.

Furthermore, research findings indicate that implementing strict environmental policies and promoting financial development can be successful strategies for addressing negative environmental impacts through the adoption of green technological advancements. Although several commitments to alleviate environmental sustainability and transition to renewable energy, several SSA countries engage in investing in non-renewable sources, particularly, in the extraction of oil and gas. To address this, policymakers and governments must implement stringent environmental rules to deter investment in non-renewable projects, foster the adoption of green innovative technologies, encourage green projects and promote energy transition for a sustainable future, and achieve SDGs 7& 13.Moreover, SSA economies should prioritize promoting financial development through the implementation of policies that aim to strengthen their financial institutions and markets. This can enhance environmental quality. Considering the impact of green innovation technologies on the environment, it is crucial to ensure that the financial sector allocates additional funds to businesses for R&D activities to adopt innovative technologies and augment awareness initiatives to educate general public of SSAregion. These policies will encourage the adoption of innovative technologies and lowering reliance on fossil fuels and thus promoting environmental sustainability. It is also important to identify potential green investment projects for the pursuit of SDGs 7 & 13.

Although several determinants of environmental quality have been analyzed, this study has some limitations. For instance, it is crucial to enhance the environmental rules for companies to guarantee that technological advancements positively impact environmental qualit[illegible] It would be intriguing to comprehend the impact of financial instit[illegible] and the development of financial markets on environmental qualit[illegible] Sub-Saharan Africa (SSA). We propose this as a potential approach [illegible] further research. Furthermore, future research has th[illegible]tential [illegible] expand this analysis by examining the relationship[illegible]we[illegible]inancial development and environmental quality in Sub-S[illegible]ran Af[illegible]n countries with British and French legal foundations. C[illegible]derin[illegible] argument in economics and finance theory [illegible]t th[illegible]ial sector[illegible] more advanced in countries with a Bri[illegible] legal syst[illegible]ompared to nations with a French legal system, i[illegible] be valuable [illegible]onduct a study from multiple perspectives i[illegible]is a[illegible] Other than [illegible]is, future research can use interaction ter[illegible] of urbaniza[illegible]nd green innovative technologies to the model f[illegible]nalyzing the po[illegible] moderating impacts.This study has empl[illegible] LCF a[illegible]proxy to mea[illegible]ure environmental quality, while future re[illegible]h ca[illegible] CO_2 emission, carbon footprints, and GHG emissions indic[illegible]easure[illegible]ironmental quality. This study focused SS[illegible]ion w[illegible]future[illegible]earch can be conducted by using the com[illegible]selecte[illegible]es in other regions to increase the genera[illegible]bility.

CRediT auth[illegible]contribu[illegible]on statement

Nudrat Fatim[illegible][illegible]riting – original draft, Methodology, Formal analysis, Conceptuali[illegible]ation. **Hu Xuhua:** Writing – review & editing, Visualization, Supervision, Conceptualization. **Muhammad Kamran Khan:** Writing – original draft, Formal analysis, Data curation. **Vishal Dagar:** Visualization, Validation, Data curation.

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Data availability

Data will be made available on request.

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